

FOOTBALL TOURNAMENT PERFORMANCE ANALYSIS

A Power BI Dashboard Analysis of FIFA WORLD CUP

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Visualization Tool: Power BI Desktop

Data Preparation: Power Query (in-model, no external DBMS)

Domain: FIFA WORLD CUP TOURNAMENT

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3. Executive Summary

This report analyzes international football tournament performance using a Power BI dashboard built on a three table data model covering 32 national teams across five tournament editions (2002–2018). The model tracks team level results, games played, wins, draws, losses, goals for, goals against, goal difference, points, and rank and surfaces them through a four page report: a landing Home page, an Executive Overview, a Team Performance page, and a Historical Trend page. Across the dataset, teams combined for 128 matches. The dashboard consistently highlights Germany as the top team across attacking output, win count, and points, and Germany posts the largest goal difference in the underlying team table. The analysis was built entirely within Power BI, data was modeled and transformed using Power Query and DAX measures rather than a separate database engine. The findings below translate these patterns into a small set of business relevant conclusions and recommendations for how the dashboard, and the underlying data collection process, should evolve.

4. Business Context & Problem Statement

Tournament organizers, federations, and sports analysts need a consolidated view of team performance that goes beyond a single tournament's results table. A recurring, comparable dataset across multiple tournament editions allows stakeholders to identify sustained strengths, emerging teams, and structural weaknesses insight that a single year scoreboard cannot provide. This analysis was framed around the following business questions:

- Which teams have the best overall attacking output across major tournaments?
- Which teams have conceded the fewest goals and hold the strongest defensive record?
- Which teams are the most successful by win count and points accumulated?
- How has scoring output trended across tournament years?
- Which teams are improving or declining in performance over successive tournament editions?
- How do win, loss, and draw ratios vary by team and by tournament year?

Power BI was selected as the tool for this project because the dataset is modest in size, relationship-driven (team and year dimensions against a single performance fact table), and best served by an interactive, filterable dashboard rather than a static report. Power BI's slicers and cross-filtering let a non-technical stakeholder explore the same questions above without writing a query.

5. Data Overview & Schema

The data model follows a star-schema pattern with one fact table and two supporting dimension tables:

- Football PB (fact table): One row per team per tournament edition, holding the performance metrics (games played, win, draw, loss, goals for, goals against, goals against per match, goal difference, points, rank) that every visual in the report is built from.

- Country (dimension table): Holds the Team name field and drives the Team slicer and axis labels across every page.
- Tournament Year (dimension table): Holds the Tournament Year field, used to drive the Tournament Year slicer and the time based axis on the trend charts on the Historical Trend page.

Football PB relates to Country on the Team field and to Tournament Year on the Tournament Year field, in each case a one-to-many relationship from the dimension to the fact table. This structure is what allows a single click on a team or year in a slicer to filter every visual on the page simultaneously.

6. Data Dictionary & Assumptions

Data Dictionary

Field	Table	Definition
Team	Country	Name of the national team / country represented in the tournament.
Tournament Year	Tournament Year / Football PB	The year in which the tournament edition took place (2002–2018).
Games Played (GP)	Football PB	Total number of matches played by the team in that tournament edition.
Win (W)	Football PB	Number of matches won by the team.
Draw (D)	Football PB	Number of matches drawn by the team.
Loss (L)	Football PB	Number of matches lost by the team.
Goals For (GF)	Football PB	Total number of goals scored by the team.
Goals Against (GA)	Football PB	Total number of goals conceded by the team.
Goals Against Per Match	Football PB	Average number of goals conceded per match played.
Goal Difference (GD)	Football PB	Goals For minus Goals Against.
Points	Football PB	Tournament points accumulated by the team.
RANK	Football PB	The team's finishing rank for that tournament edition.

Key DAX Measures

Measure	Purpose
Total Countries	Distinct count of teams represented in the filtered dataset.
Total Matches	Sum of games played across all teams and tournament editions in context.
Total Goals Scored	Sum of Goals For across the filtered dataset.
Average Goal Scored	Average goals scored per match in the filtered dataset.
Highest Goal Difference	Maximum Goal Difference value in the filtered dataset.
Highest Point	Maximum Points value in the filtered dataset.
Most Win	Team with the highest Win count in the filtered dataset.
Best Attack	Team with the highest Goals For in the filtered dataset.
Best Defense	Team with the lowest Goals Against in the filtered dataset.
Most Prolific Team	Team with the highest cumulative scoring measure across editions.

Assumptions

- Each row in Football PB represents a team's aggregated record for a single tournament edition, not an individual match, the model does not support match by match or head to head analysis.
- Points follow a standard football scoring convention (win/draw/loss weighting); the exact weighting formula was not independently verified against the source data in this review.
- Tournament Year is treated as a discrete category by the model (used for slicing and axis grouping), not as a continuous date field.
- "Best Attack," "Best Defense," and "Most Prolific Team" are single team superlative measures, they surface one leading team per filter context and are sensitive to whatever Team/Year slicer selection is active.

7. Data Quality Assessment

Before building the Power BI data model and dashboard, a comprehensive data assessment and quality check was carried out to ensure the dataset was complete, accurate, and suitable for analysis. Five FIFA World Cup Excel files (2002, 2006, 2010, 2014, and 2018) were imported into Power Query and combined into a single fact table. Rather than manually entering tournament years, a Tournament Year column was derived automatically from the source file names, ensuring consistency and reducing the risk of human error.

The dataset was then inspected for data quality issues. Column data types were reviewed and corrected, converting text fields to numbers where appropriate to enable accurate calculations and aggregations. The first row of each dataset was promoted to headers, and column names were standardized across all files to ensure successful appending and modelling.

A validation process was performed to confirm that all five tournament files were successfully combined. Row counts from the combined table were compared with the individual source files to verify that no records were lost or duplicated during the transformation process. Key football metrics, including matches played, goals scored, goals conceded, wins, draws, and losses, were also reviewed against the original datasets to ensure data integrity.

Duplicate checks were conducted on the Country/Team field before creating the Team dimension table. Distinct values were extracted to eliminate repeated country names and establish a clean lookup table for the star schema. A separate Year dimension was also created from the Tournament Year field to support efficient filtering and time-based analysis.

Missing values and inconsistencies were examined throughout the dataset. No significant missing records affecting the project objectives were identified. Team names were checked for spelling consistency to prevent duplicate representations of the same country, while numeric fields were verified to ensure they contained valid numerical values rather than text.

After completing these quality assurance procedures, the data model was structured into a star schema consisting of one central Fact Table and supporting Country/Team and Year

dimension tables. Relationships were validated to ensure correct filter propagation across visuals.

These data assessment and quality checks ensured that the dataset was accurate, consistent, and reliable for building interactive dashboards, calculating KPIs, and performing comparative analysis of FIFA World Cup team performances across multiple tournament years.

8. Methodology

Data Preparation & Modeling Method

1. Imported the tournament performance dataset into Power BI Desktop via Power Query.
2. Built a star-schema data model: the Football PB fact table related to two dimension tables, Country and Tournament Year.
3. Set relationship cardinality to one-to-many from each dimension table to the fact table, enabling cross-filtering from slicers to every visual on a page.
4. Created DAX measures for the five Executive Overview KPIs: Total Countries, Total Matches, Total Goals Scored, Average Goal Scored, and Highest Goal Difference.
5. Created DAX superlative measures for the Team Performance page: Best Attack, Best Defense, Most Prolific Team, Most Win, and Highest Point.
6. Validated the headline KPI cards against the detailed team table (RANK, Team, GP, W, D, L, GF, GA, GD, Points) to confirm the summary and detail levels were consistent.
7. Segmented the report into four pages, each targeting a different stakeholder question (see Visualization Method).

Visualization Method

1. Home Page: A branded landing page with navigation buttons into the three analytical pages.
2. Executive Overview: Five KPI cards (Total Countries, Total Matches, Total Goals Scored, Average Goal Scored, Highest Goal Difference), a Team and Tournament Year slicer, a Goal Scoring Trend line chart, a full Team Performance table, and a Match Outcome Distribution 100% stacked column chart by tournament year.
3. Team Performance: Five KPI cards (Best Attack, Best Defense, Most Prolific Team, Most Win, Highest Point), a Goal Scoring Ranking bar chart, a Goal Conceded Ranking bar chart, a Best Overall Performers by Goal Difference line chart, a Win-Loss Ratio 100%-stacked bar chart, and a supporting table, with Team and Tournament Year slicers.
4. Historical Trend: Six line charts trending goals scored, wins, goals conceded, attacking improvement, team ranking, and top-team performance improvement across tournament years.

9. Analysis & Findings

Across the full, unfiltered dataset the dashboard reports 32 competing countries and 128 total matches every tournament year, from 2002 to 2018 of World Cup years Total of 59 countries participated in the Tournament, 640 matches were played, 14 is the Highest Goal Difference a team had during that period, 1.24 Average Goal Scored and 793 Goals Scored are recorded. The sub-sections below translate the patterns visible across the Team Performance and Historical Trend pages into business relevant conclusions.

Attacking Performance

The Best Attack, Most Prolific Team, and Most Win measures on the Team Performance page consistently changes across tournament years but Germany slightly appeared twice which shows the attacking threat they possessed and it shows in the dashboard they have the best attack ranking across all tournament years reviewed. For a federation or broadcaster, this signals a team whose commercial and competitive profile is worth building marketing and scouting narratives around; for a rival federation, it flags the benchmark attacking output a competing squad needs to plan against.



Defensive Performance

The Data given is not enough to accurately review team with the best defense. However, the team with best Defense consistently varies across tournament years with the likes of Germany, Costa Rica, Switzerland, Portugal and Denmark showing up in the dashboard. Combined with the Goal Conceded Ranking chart, this points to a small group of teams which is a useful signal for talent scouts and opposition analysts alike.

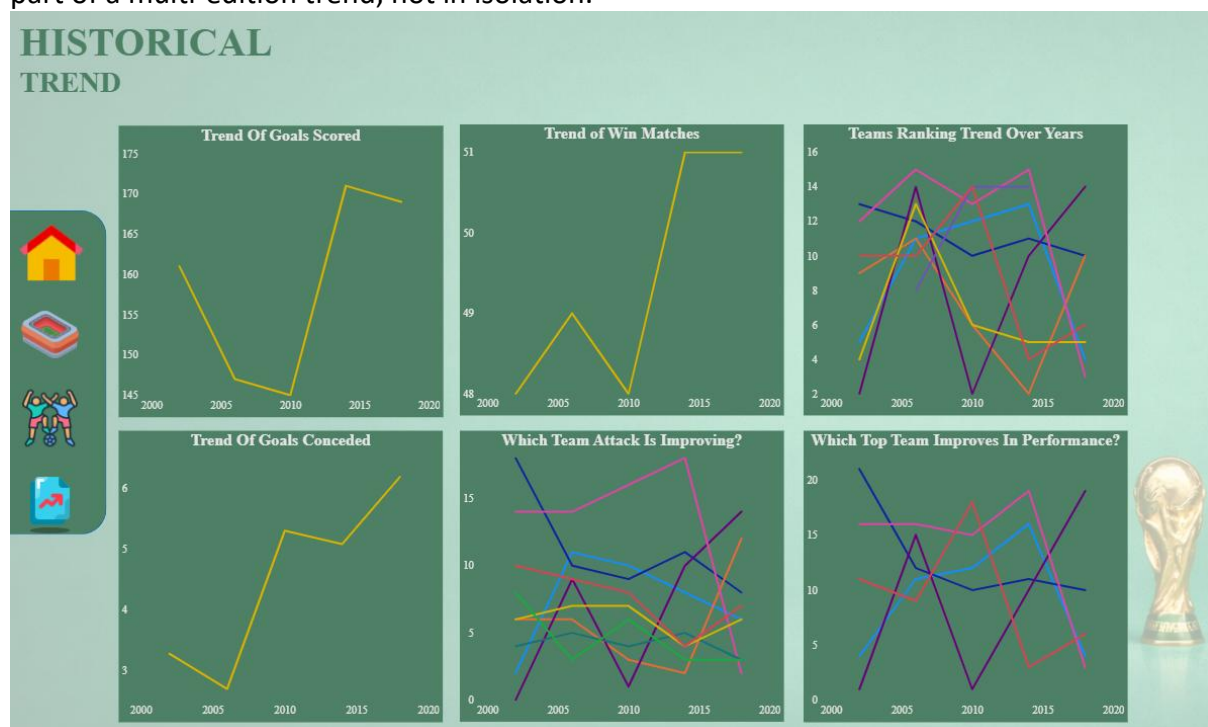
Overall Team Success

The Win-Loss Ratio by Team chart and the underlying team table show a clear front tier of teams led by Germany and Brazil in the table views reviewed, with Germany posting the largest goal difference recorded in the table, separated from a longer tail of teams with more even win/loss splits. This two-tier pattern is typical of tournament football and is useful for setting realistic seeding or investment expectations for mid-tier teams.



Tournament-Year Trends

The Goal Scoring Trend Across Tournament Year and Trend of Goals Scored charts track total goals scored across the five tournament editions in the dataset (2002–2018). Fluctuation across editions, rather than a single sustained direction, is the dominant pattern a reminder that single tournament scoring totals are a noisy metric and should be read as part of a multi-edition trend, not in isolation.



Team Trajectory / Improvement

The Historical Trend page's "Which Team Attack Is Improving?" and "Which Top Team Improves In Performance?" charts are purpose-built to answer a question the year-by-year totals alone cannot: not just who is best right now, but who is trending upward. This is the most forward-looking view in the dashboard and is the one most directly useful for scouting and long-range planning decisions.

10. Data Limitations & Business Impact

The data model is built at the team per tournament edition grain, which structurally limits the questions this dashboard can answer:

- No match by match or opponent level data: Questions like "how does Team A perform specifically against Team B" or "is a team stronger in group-stage vs. knockout matches" cannot be answered.
- Only five tournament editions (2002–2018) are captured: Trend and improvement charts are working with a short time series, which limits statistical confidence in any long-run trajectory claim and excludes more recent tournaments.
- No player level data: Individual contribution to a team's attacking or defensive record (e.g., which player is driving a team's goal-scoring output) cannot be analyzed from this model.
- No match date/stage field: Within tournament questions (form across the group stage versus the knockout rounds) are out of scope for the current model.

11. Key Insights

- A small, consistent group of teams led by Germany on attacking and results measures for a disproportionate share of the dashboard's headline "best" superlatives.
- Germany's goal difference stands out as the largest recorded in the team table, suggesting the strongest combined attack and defense balance among the teams reviewed.
- Goal scoring output fluctuates across tournament editions rather than trending in one sustained direction, so single tournament totals should be read in the context of the multi-edition trend.
- The dashboard is architecturally ready to answer "who is improving" questions, which is typically more valuable for forward planning than static, single-edition leaderboards.
- The current data grain (team-per-tournament) is the dashboard's main structural constraint, and any expansion of scope should prioritize match-level granularity over additional KPIs at the current grain.

12. Recommendations

1. Confirm and freeze a single unfiltered baseline for every headline KPI before external distribution, because several cards are filter-sensitive and different dashboard views produced different figures for the same card during development.
2. Extend the data model to match-level granularity where the source data allows it, because opponent specific and stage specific (group vs. knockout) analysis is currently structurally impossible.
3. Add tournament editions beyond 2018 as they become available, because a five-point time series limits confidence in any "improving vs. declining" trend claim.
4. Document the exact points scoring convention used in the Points field, because this report's methodology currently assumes a standard win/draw/loss weighting that was not independently verified.
5. Add a short on page legend or tooltip explaining how the single team superlative measures (Best Attack, Best Defense, Most Prolific Team) behave under multi-team slicer selections, because these measures are easy for a non-technical viewer to misread as an average or ranking rather than a single team maximum.
6. Pair the Historical Trend page's improvement charts with a written scouting note once match-level data is available, because the trend lines currently show *that* a team is improving but not the underlying cause.

13. Limitations & Assumptions

The historical data in this model can describe what happened across the five captured tournament editions and can support relative comparisons between teams and between editions. It cannot, on its own, predict future tournament outcomes, account for squad changes, injuries, managerial changes, or other external factors that are not represented as fields in the model. Any forward-looking statement drawn from the Historical Trend page's improvement charts should be treated as a description of past trajectory, not a forecast, until external context is layered on top of it.

14. Appendix

- A. Data model relationships — Football PB (fact) to Country (dimension, on Team) and to Tournament Year (dimension, on Tournament Year), both one-to-many.
- B. DAX measures — Total Countries, Total Matches, Total Goals Scored, Average Goal Scored, Highest Goal Difference, Highest Point, Most Win, Best Attack, Best Defense, Most Prolific Team.
- C. Validation checks performed — headline KPI cross-check (Total Goals Scored ÷ Total Matches ≈ Average Goal Scored) on the Executive Overview page.
- D. Dashboard build artifacts — four report pages (Home Page, Executive Overview, Team Performance, Historical Trend) comprising 24–25 visuals per analytical page, referenced throughout Section 9 by chart name.